

1. $L = 350 \mu\text{m}$, $n = 3.4$

Estimate phase oscillation using initial wavelength of $1.55 \mu\text{m}$

$$m\lambda = 2nL \rightarrow m = \frac{2nL}{\lambda} = \frac{2(3.4)(350 \times 10^{-6})}{1.55 \times 10^{-6}} = 1535.48$$

$$\underline{m = 1535}$$

spacing between modes:

Make guess: $m_1 = 1535 \rightarrow 2L = m_1 \left(\frac{\lambda_1}{n} \right) \quad \lambda_1 = \frac{2Ln}{m_1}$

$m_2 = 1536 \rightarrow \lambda_2 = \frac{2Ln}{m_2}$

$$\Delta\lambda = \lambda_1 - \lambda_2 = 1 \times 10^{-9} = \underline{1 \text{ nm}}$$

2. delay: $\frac{\Delta\tau}{L} = \frac{(n_1 - n_2)^2}{8cn_1}$

a. $n_1 = 1.5$, $n_2 = 1.485$, $L = 40 \text{ km}$

Max bit rate $R_b \leq \frac{1}{4\Delta\tau} \rightarrow \Delta\tau = \frac{(1.5 - 1.485)^2}{8 \cdot 3 \times 10^8 \cdot 1.5} \cdot 40 \times 10^3 \text{ m}$

$$\underline{R_b = 100 \text{ Mbps}}$$

b. $\Delta\tau = \frac{(1.49 - 1.48)^2}{8 \cdot 3 \times 10^8 \cdot 1.49} \cdot 400 \times 10^3 \rightarrow R_b \leq \underline{22.35 \text{ Mbps}}$

3. a. $L = 100 \text{ km} \rightarrow$ total attenuation: $100 \text{ km} \times 0.22 \text{ dB/km} = 22 \text{ dB}$

Power = $22 \text{ dB} + 1 \text{ dBm} = \underline{23 \text{ dBm}}$

$L = 200 \text{ km} \rightarrow$ total attenuation: 44 dB

Power = $44 \text{ dB} + 1 \text{ dBm} = \underline{45 \text{ dBm}}$

$L = 300 \text{ km} \rightarrow$ total attenuation = 66 dB

Power = $66 \text{ dB} + 1 \text{ dBm} = \underline{67 \text{ dBm}}$

$L = 400 \text{ km} \rightarrow$ Power = $88 \text{ dB} + 1 \text{ dBm} = \underline{89 \text{ dB}}$

b. splice loss: $9 \text{ splices} \times \frac{0.25 \text{ dB/loss}}{\text{splice}} = 2.25 \text{ dB}$

$\rightarrow$ total loss = $44 \text{ dB} + 2.25 \text{ dB} + 1 \text{ dB} = 47.25 \text{ dB}$

connector loss: $2 \text{ connectors} \times \frac{0.5 \text{ dB}}{\text{connector}} = 1 \text{ dB}$

Power = $47.25 \text{ dB} + 1 \text{ dBm} = \underline{48.25 \text{ dBm}}$

4. a. $D = 16 \text{ ps/nm/km}$, $L = 500 \text{ km}$, $\lambda = 1550 \text{ nm}$, $\Delta\lambda = 0.1 \text{ nm}$

$$\Delta T = D \times \Delta\lambda \times L = 16 \text{ ps/nm/km} \times 500 \text{ km} \times 0.1 \text{ nm} = 80 \text{ ps}$$

$$R_b \leq \frac{1}{4\Delta T} = \underline{3.12 \text{ Gbps}}$$

b. $L \leq \frac{1}{4 \times D \times \Delta\lambda \times R_b} = \frac{1}{4 \times 16 \text{ ps/nm/km} \times 0.08 \text{ nm} \times 40 \text{ Gbps}} = \underline{4.88 \text{ km}}$

5. $\Delta T = 17 \text{ ps/km/nm} \times 100 \text{ km} \times 10 \text{ nm} = \underline{17 \text{ ns}}$